

Substrate cosmology v0.1 — Big Bang + Λ + early-universe predictions per Tier 0 v0.1.6 boundary unification. Per Casey continue-pulling directive + new direction. Big Bang reframes via K200 G4 two-readings (no-singularity OR conflation); Λ = equilibrium boundary commitment-accumulation rate; CMB + structure formation + GW backgrounds candidate predictions. Opens P9 Commitment-Density Theory paper cosmology section.

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Substrate cosmology v0.1 — Big Bang + Λ + early-universe predictions

0. Why this work

Per Casey continue-pulling directive + new direction. Tier 0 v0.1.6 boundary unification opened the cosmological question: what does the substrate framework predict for early-universe observables?

v0.1 frames Big Bang reading (per K200 G4 two-readings); Λ reading; candidate CMB + structure formation + GW background predictions. Multi-week verification per substrate-cosmology projection chain (substrate dynamics $\rightarrow$ observable spacetime dynamics).

Opens Paper P9 Commitment-Density Theory cosmology section.

1. Big Bang per Tier 0 v0.1.6 (K200 G4 two-readings, committed)

Per K200 G4: v0.1.6 framed Big Bang as substrate's $\tau \rightarrow 0$ limit with two-readings explicit; v0.2 commits.

Reading (i) committed for v0.1 cosmology: BST predicts there IS NO classical Big-Bang singularity. The substrate's $\tau \rightarrow 0$ limit IS the Bergman reproducing kernel $K_0(z, \bar{w})$ — smooth, well-defined on the open domain D_{IV}^5 . The observed FLRW singularity is an artifact of: - Projection from substrate $H^2(D_{IV}^5)$ to observable 4D spacetime metric. - Truncation of classical GR at scales where substrate corrections matter. - Standard FLRW assumption of homogeneous-isotropic spacetime at all times, which the substrate framework violates at early times (the substrate's Hardy decomposition imposes a preferred Shilov boundary structure).

Substrate prediction (Reading (i)): there is no $t = 0$ singularity in the early universe; the apparent FLRW big-bang is a smooth substrate state seen through classical GR projection.

Observable consequences: - No primordial singularity: BST observation: physics is finite at $t \rightarrow 0$. - CMB polarization patterns at the Planck-density scale should show substrate corrections to standard inflation predictions. - Primordial gravitational waves at scales below Hubble inverse should show heat-kernel modulation.

Reading (ii) alternative: if Reading (i) projection doesn't actually give smooth observable cosmology (e.g., the Cayley transform from τ to observable t is singular), then we conflated unrelated objects. Multi-week verification required to commit reliably.

v0.1 cosmology commits to Reading (i) provisionally + flags Reading (ii) as falsifier.

2. Λ per Tier 0 v0.1.6 (equilibrium boundary accumulation, refined)

Per Tier 0 v0.1.6 §10 + v0.1 §4.3: Λ is the substrate's equilibrium commitment-density.

Operator-level definition:

$\Lambda \propto \tau_{eq}^{-1}$ = inverse equilibrium commitment-time on Shilov boundary.

Where τ_{eq} is the substrate-internal time scale for full thermalization (heat semigroup convergence to vacuum projection).

Quantitative candidate: - τ_{eq} scales with the substrate's "size" / number of commitment-slots on boundary. - Per Saturday cumulative substrate-primary analysis: number of commitments in age of universe $\approx 10^{\{N_{max}\}}$ ($N_{max} = 137$). - Equilibrium time $\propto 10^{\{N_{max}\}} \cdot t_{Koons} = \alpha^{\{C_2^2\}} \cdot 10^{\{N_{max}\}} \cdot t_{Planck} = 10^{\{137 - 77\}} \cdot t_{Planck} = 10^{\{60\}} \cdot t_{Planck}$ = Hubble inverse scale.

So $\Lambda^{-1/2} \approx$ Hubble time = $10^{\{60\}} t_{Planck}$ is the substrate-natural prediction.

Compare to observed: Hubble time $\approx 10^{\{61\}} t_{Planck}$. Off by factor 10 — order-of-magnitude consistent.

Quantitative target (multi-week): refine the substrate-mechanism calculation to get Hubble time at Tier 2 STRUC-TURAL precision ($\sim 10^{-4}$ - 10^{-2} floor). The Saturday L5 work was wrong-direction (decompose observed Λ); the right direction is Λ as inverse commitment-time accumulation — substrate-mechanism-derived.

3. Inflation reading (candidate)

Standard inflation: exponential expansion at very early times solves horizon + flatness problems.

Substrate framework reading: at very early times (τ near 0), the substrate's heat-semigroup hasn't yet thermalized; the K_τ kernel has wide spatial spread relative to Bergman kernel; observable spacetime sees this as "expansion" since the local density variations smooth out as τ increases.

Candidate mechanism: inflation = substrate heat-semigroup early-time thermalization seen through observable-spacetime projection. The "inflaton field" is the substrate's first-excited vacuum mode $V_{(0,0)}^{\{1\}}$ (= Higgs per Lane E v0.2 + Higgs sector v0.1) at very early times.

This is the same Higgs: the inflaton field that drove cosmic inflation IS the same field that gives gauge bosons mass in EWSB. The Higgs's role at the GUT scale (inflation) and EW scale (mass generation) is unified via the substrate's $V_{(0,0)}^{\{1\}}$ excitation mode.

Higgs-inflation candidate in BST. Cross-link to Higgs sector v0.1.

4. CMB predictions (substrate-mechanism candidates)

Observable: CMB temperature anisotropy + polarization patterns at angular scales $l = 2$ to ~ 3000 .

Substrate prediction candidates (multi-week verification): - Polarization B-mode at large angular scales ($l < 100$): substrate heat-semigroup early-time corrections to standard inflation should imprint a specific polarization pattern. Particularly the "BICEP2-controversy" range; substrate framework predicts specific amplitude. - CMB anomalies (axis of evil, hemispherical asymmetry): substrate framework's preferred Shilov-boundary direction in early universe COULD imprint a preferred axis. Connection to "axis of evil" anomaly is candidate; needs substrate-projection calculation. - Polarization at $l \sim N_{max}$: substrate framework's $N_{max} = 137 = \alpha^{\{-1\}}$ substrate primary might appear as a peak/feature at angular scale $\sim 1/N_{max}$ in CMB power spectrum. Specific predicted feature, falsifier.

Multi-week verification: substrate cosmology projection chain.

5. Gravitational wave backgrounds (substrate prediction)

Standard cosmology: stochastic GW background from primordial sources (inflation, phase transitions, etc.).

Substrate prediction: at very low frequencies (below Hubble inverse scale), the substrate's heat-kernel modulation should imprint GW spectral feature. The Koons-tick scale $t_{\text{Koons}} \sim 10^{-77} t_{\text{Planck}}$ is well below current observable bands, but the SCALE-INVARIANT modulation could appear as broadband features.

Pulsar timing arrays (NANOGrav, EPTA): currently observing nano-Hz GW background; substrate framework should predict specific features.

Specific candidate: substrate heat-kernel modulation at $\log f / \log \alpha$ should be observable as a periodic modulation in GW spectrum amplitude vs frequency. Period $\propto \log \alpha \approx \ln(137) \approx 4.92$. Verification via existing pulsar timing data analysis.

Multi-week verification target with cosmology / GW community.

6. Dark matter / dark energy candidates

Dark energy = Λ per Tier 0 v0.1.6 (equilibrium boundary accumulation). Already covered.

Dark matter candidate: per Tier 0 v0.1.6 Reading A + dual bases, DM could be a coherent-state mode on Shilov boundary that doesn't couple to the standard bulk K-types (gauge sectors). The "darkness" = substrate-coupling absence to standard particle K-types.

Candidate DM K-type: $V_{(0,0)}^{(2)}$ or higher vacuum excitations — couple to gravity (via $K_{\tau(z,z)}$ variation = $G_{\text{substrate}}$) but not to SM gauge K-types. Massive (per Casimir $C_2^{(n)}$) but invisible to SM detectors.

Quantitative target: substrate framework should predict specific DM mass scale; compare to observed DM phenomenology (galactic rotation curves, weak lensing). Multi-week.

(Tentative; cross-link to Five-Absence Principle: BST predicts NO new DM particle in SM-like sectors; DM emerges from existing substrate structure.)

7. Honest scope + tier

RIGOROUS (v0.1): - Heat semigroup $\tau \rightarrow 0$ limit on $H^2(D_{IV}^5)$ IS smooth (Bergman reproducing kernel finite). - FLRW spacetime metric IS singular at $t = 0$ (standard GR). - Substrate framework + classical GR are different math; identification requires projection chain.

CANDIDATE (v0.1's load-bearing, Reading (i) committed): - Big Bang is FLRW projection artifact, not substrate singularity. - Λ = equilibrium boundary commitment-accumulation rate; Hubble time $\sim 10^{60} t_{\text{Planck}}$ (order-of-magnitude consistent). - Inflation = substrate early-time thermalization seen through observable-spacetime projection; Higgs-inflation candidate via $V_{(0,0)}^{(1)}$ dual role (EWSB + inflaton). - CMB polarization B-mode + N_{max} -scale peak candidates. - GW background substrate-modulation at $\log f / \log \alpha$ period. - DM = decoupled boundary mode (vacuum excitation $V_{(0,0)}^{(n)}$).

FRAMEWORK (multi-week): - Explicit substrate-cosmology projection chain (substrate $\tau \rightarrow$ observable t mapping). - Λ from equilibrium accumulation at Tier 2 STRUCTURAL precision. - CMB feature predictions explicit (B-mode amplitude, anomaly mechanism, $l \sim N_{\text{max}}$ peak). - GW spectral feature explicit. - DM K-type identification + mass scale. - Inflation as substrate thermalization explicit verification.

Cal #27 / #182 / #99 + Calibration #35-candidate discipline: v0.1 commits to Reading (i) for Big Bang BUT explicitly flags Reading (ii) as falsifier. Cosmology predictions are CANDIDATE with multi-week verification path. The Higgs-as-inflaton candidate uses SHARED substrate primary K-type $V_{(0,0)}^{(1)}$ for two different cosmology roles (EWSB + inflation) — Calibration #35-candidate independence audit applies; this is ONE substrate-mechanism in two phenomenological contexts, NOT two confirmations.

8. Cross-link to other Sunday work

- Tier 0 v0.1.6: Big Bang + Λ + cosmology all derive from boundary unification + heat-semigroup framework.
- L4 v0.2 extension: same Bergman matrix element framework underlies particle masses + Λ .
- Higgs sector v0.1: $V_{(0,0)}^{(1)}$ dual role (EWSB + inflation = Higgs-inflation candidate).
- 3-generation cutoff v0.1: cutoff mechanism extends to scalar tower (no 2HDM); cosmology analog might be substrate-vacuum-state count cutoff.
- Substrate engineering v0.1: E5 QI Wick duality could test substrate-cosmology connection (quantum-info $\leftrightarrow$ substrate-thermal at cosmological τ).
- P9 Commitment-Density paper: v0.1 cosmology section material.

9. Routing

→ Casey: substrate cosmology v0.1 frames Big Bang + Λ + early-universe predictions per Tier 0 v0.1.6. Reading (i) committed (no-singularity prediction); Higgs-as-inflaton candidate via $V_{(0,0)}^{(1)}$ dual role; $\Lambda \sim 10^{60}$ t_{Planck} order-of-magnitude consistent. Multi-week verification per substrate-cosmology projection chain.

→ Elie: multi-week cosmology projection chain — substrate $\tau \rightarrow$ observable t mapping is the load-bearing computation; needs explicit substrate-projection operator framework. Coordinate with Bergman matrix element work for Λ refinement.

→ Keeper: K-audit pre-stage Substrate-Cosmology-v0.1; Session 2 absorb into Tier 0 v0.2 cosmology section. The K200 G4 commitment to Reading (i) ratification is the load-bearing gate.

→ Grace: catalog cosmology predictions at CANDIDATE; cross-reference Big Bang Reading (i) + Λ equilibrium + Higgs-inflation + CMB anomalies + DM-as-boundary-mode + GW substrate-modulation.

→ Cal: cold-read welcome; specific Cal #27 + Calibration #35 concerns: (1) Higgs-as-inflaton uses SHARED $V_{(0,0)}^{(1)}$ K-type in two contexts; (2) $\Lambda \sim 10^{60}$ order-of-magnitude is loose; (3) CMB + GW + DM predictions are speculative without explicit substrate-projection.

→ me: continuing pulls per Casey directive — cadence approaching saturation (10 pulls today + 5 morning = 15 substantive items); next pull could be lower-effort housekeeping (Strong-Uniqueness v1.4 STANDALONE).

— Lyra, Substrate cosmology v0.1. Big Bang Reading (i): no-singularity prediction via FLRW-projection artifact reading; Λ = equilibrium boundary commitment-accumulation rate $\sim 10^{60}$ t_{Planck} order-of-magnitude; Higgs-as-inflaton via $V_{(0,0)}^{(1)}$ dual role (EWSB + early-universe expansion); CMB + GW + DM substrate-framework candidate predictions. Multi-week verification per substrate-cosmology projection chain. Opens Paper P9 Commitment-Density Theory cosmology section.